

Big Data Storage and Analysis for Smart Farming

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Abstract—Smart Farming has always been referred to as agriculture, but nowadays, that is no longer the case. Today, Smart farming is made up of Precision Agriculture (PA) and Precision Livestock Farming (PLF). Big Data technologies and algorithms can be relevant for managing and monitoring data related to any farm. Precision livestock farming concerns genetics, animal welfare, animal nutrition, reproduction, species protection and animal health. This paper presents a general overview of Big Data tools that can be applied in a smart farming application. New Technologies are offering many tools used to facilitate the management of data collection, risk minimization, climate change anticipation, secure storage and analysis, etc. The main purpose of Big Data tools is to increase productions in order to offer higher quantities while ensuring higher quality products. However, they remain some issues that need to be accomplished.

Index Terms—Smart Farming, Big Data, Precision Agriculture, Precision Livestock Farming.

INTRODUCTION

The world's population is growing exponentially while resources are being depleted at an ever-increasing rate. Several studies agree that the population will be close to 9.6 billion by 2050 [1]. According to the United Nations (UN), if this proven to be true, the equivalent of nearly three planets (like earth) could be needed to provide necessary resources to maintain the current lifestyles. Already today (according to the UN) one person out of nine in the world is undernourished (815 million) - the equivalent of 11% of the world's population-. Hence, malnutrition and hunger can be seen as a global problem. For some years now, scientific research has been carried out to address this problem. This research is mainly related to agriculture and livestock and its goal is to ensure healthy and sustainable food. Those studies has been published under several names: Smart Farming (SF), Sustainable Farming (Agriculture), Precision Agriculture (PA), etc. In this study, we will refer to these concepts as Smart Farming. Smart Farming is a global initiative that aims to preserve resources and to ensure both sustainable and healthy agriculture and livestock with the help of advanced technologies-. In recent years, new technologies have drawn many people's attention, regardless of their industrial sector. Technologies such as Big

Data (BD), Internet of Things (IoT), Cloud Computing (CC), Geographical Information Systems (GIS), Control Stations and more are being used to their full potential and have proven to be increasingly useful and vital. The agricultural activity in the 21st century is driven to a greater or lesser extent by a large mass of data. Data are coming from many different sources such as sensors, weather stations, biodiversity, soils, climate, etc. These data are also intended for various uses such as: animal genetics, animal behavior, animal feed, agricultural production, species protection, breeding or reproduction, etc. The exploitation of these data requires adapted technologies to fit the activity sector constraints, the usage constraints and the legal constraints related to it. Big Data technologies allow, among other things, to obtain relevant and important information from the collected data in order to improve the activity. Our work is particularly focused on cattle breeding.

The aim of this work is to present Big Data technologies and tools for data storage and analysis. Thus, our goal is to understand how they work, the benefits they offer, understand the complexity and the need to use them together. We will also present Smart Farming and how Big Data is revolutionizing this sector in many different ways. To do that, a bibliography study was conducted in three steps: (1) collect related works, (2) spotting out the relevant work and finally (3) resume and analyze the direct implied work.

In the same way, we filtered papers, selecting those referred to Big Data applied on Smart farming, livestock farming, and other terms used to name Smart farming and especially livestock farming. We also look about some project reports or web site related to the same subject or covering some project alike and some review papers. We closed this section by resuming the selected papers.

This paper is organized as follows: section I presents the work done related to Big Data and Smart farming, while section II presents a general view of Big by describing the evolution of dimensions of Big Data and then by illustrating the Big Data Value Chain. The general overview of Smart farming is presented in section III in which we present all the fields of smart farming with respect to the different sectors. Then, we present the use of Big Data in Smart farming.

Here, we present Dataflow on smart farming platform and an architecture for data analysis. Finally, we ended our article with a conclusion and a little description of what can be done in the future.

I. RELATED WORKS

There exist a lot of scientific papers related to smart farming. Many papers speak about Precision agriculture but, some of them also talk about precision livestock. Schönfel and al. [2] show how Big Data improve crops productivity. The paper presents the importance of Big Data in the increase of the world food production by showing how technologies can help farmers and stakeholder in the agricultural sector. Even though technologies have many advantages when being used in smart farming and precision agriculture, some issues remain, among others, such as the data protection and data ownership rules of the new GDPR (General Data Protection Regulation) convention in force in the EU, as well as IT security issues. Kamilaris and al. [3] resume the “Big Data analysis in agriculture” with the purpose of presenting how Big Data can be applied in all the different fields of agriculture. The authors first presented Big Data then later on, they spoke about the different areas in which Big Data analysis is applied in agriculture. Furthermore, they have presented some agricultural sector where Big Data is not yet applied but will be welcome. Alreshidi Eissa [4] has presented a Smart Sustainable Agriculture (SSA) platform coupled with IoT and AI technologies. Thus, showing how new (high) technologies (AI, IoT, CC, robotic, etc.) enhance agriculture in all (its various) domains i.e. Human resources, Crops, Weather, Soil, Livestock, Irrigation/Water, Fertilization, Pests, etc. The papers begins with a description of need, challenges and urgency. Moreover, other works closed to Smart Sustainable Agriculture (SSA) have been studied and a layered architecture divided in seven layers recovering all needs has been proposed. Cambra Baseca and al. [5] have presented a Smart Farming software called PLATEM. The authors have presented the existing Precision Agriculture system. This was done by presenting the Smart Farming data management cycle, architecture and the architectural view of the smart decision system.

Regarding livestock farming, Sun He and al. [6] has presented one of the major aspects of animal behavior analysis by image processing. The authors used specifically image acquisition, image processing, image analysis and classification algorithm to make decisions. That process is used to predict animal growth (in this case animals are pigs). Linear regression is used for prediction and real-time condition monitoring. Ivan Andonovic and al. [7] has shown how new technologies can be used to control animal activities (in particular eating/ruminating), to enhance animal welfare, to ensure fertility, reproduction and control the pregnancy rate. Also to detect reproduction diseases and poor oestrus.

The aim of our Work is to put forward a new view of Big Data tools by providing a simple but robust architecture for PLF system. Simple with respect to the installation, deployment and usage. Robust for stable and scalable.

II. GENERAL OVERVIEW OF BIG DATA

Nowadays, “Big Data” can be define in several different ways. “**Big Data**” refers to a volume of data in the order of exabytes and much more [8] [9]. All the different definitions of Big Data can be summarized in a criteria which is also known as the Vs of Big Data or Dimension of Big Data [10].

Dimensions of Big Data

Big Data is characterized by some Vs words referred to as its dimensions. Initially, three of these words (volume, velocity, variety), were known to be the main ones. However, more and more have been added, then they were five main words, and later ten which is why today we speak of the 10V’s of Big Data [11] as shown by table I.

TABLE I
DIMENSION OF BIG DATA. FROM [12]

Big Data Vs	Description
1. Velocity	the speed at which data are generated and transmitted
2. Volume	Size of data generated, treated and transmitted
3. Variety	Types and models representing data (formats)
4. Veracity	Data accuracy and trustworthy
5. Value	Usable, useful and important data
6. Venue	Distinguish heterogeneous data from various sources
7. Validity	Quality, governance, master and management data
8. Variability	Dynamic, evolving behavior in data sources
9. Vagueness	Confusion over meaning of data
10. Vocabulary	Models and semantics that describe and structure data

Big Data Value Chain

Big Data Value Chain is described by the following steps:

Data Acquisition: This step contains all the processes which are use to collect, filter and clean data that are coming from any source and transmitted to a storage system, [13]. Big Data improves acquirement to enhance high volume of various Data in higher speed build in a distributed system (environment). So, some required infrastructures are needed.

Data Analysis: Here, we are dealing with how data are treated. This process involves exploring, transforming, modeling and de-noising data for a future use and decision-making. This step helps to enhance the accuracy of data and distinguish those which are usable, useful or important with respect to their needs or to their trustworthiness. It is also relevant for extracting information from data and gaining knowledge from information which are known to be very important from a business point of view [14].

Data Curation: stands for “process used to clean, to purify and to enhance data before they are used”. The main goal of this process is to increase data quality [15].

Data Storage: refers to a process that aims to save data in a secure system for any future use and needs. Data storage system has evolved during the last years, and with the rising of the Big Data, it is becoming a challenge for the community [16]. Storage systems are also related to the nature of data.

Indeed, data are either structured or not and you can choose the relevant storage system [17].

Data Usage: This aim of this process is to provide data for all needs, specifically for user (customer), stakeholders, etc. This is the view side which is also called a Dashboard. It can be a web, mobile or desktop application.

III. GENERAL OVERVIEW OF SMART FARMING

We can notice that major scientific works related to Smart farming generally talk about agriculture [2], [5], [18]–[22], etc. Smart farming encompasses many aspects other than growing crops such as animal husbandry and many other areas (beekeeping, etc.).

Five specific fields are commonly covered by Sustainable Agriculture (SA), Smart Sustainable Agriculture (SSA), and Precision Agriculture (PA) in agriculture [4]:

- Soil is the major aspect in an agriculture. It provides nutrients that are absorbed up by crops and then transmitted to animals and humans. The main problem here is that soil's health deprecated with time. Healthy soil depends on three main factors: geography, climate and exploitation.
- Crops are the main outcome of agriculture. The substance that provide profit to owner and other partners. It can be separated into four groups [4]: *food crops* for human consumption, *feed crops* for animal (livestock) consumption, *fiber crops* for cordage or textiles and finally *oil crops* for industrial needs/uses.
- Machines include any material, equipment or any kind of machinery equipment used to assist farmers in their work. In SSA/PA, machines should be fast and intelligent, less pollutant (produce less CO₂ emissions) and energy less.
- Irrigation and Water: The challenge faced in the SSA/PA is to increase food production while reducing wasted water. Various techniques exist to optimize the use water in agriculture.
- Pests and Fertilization is focusing on chemical products used to ensure production. The purpose is to secure production against harmful animals that can damage, destroy or eat crops (pests). Moreover, they help crops grow fast, produce more/better (fertilizer). However, if they are used a lot, they can cause many damages on environment and consumer (humans and animals).

For Livestock farming or Precision Livestock Farming we can consider five categories [23]:

- Animal behavior regards the habit of animals during their live in their environment, anticipates anxiety and some diseases. This can help farmers to manage, control and administrate the farm in a more efficient manner.
- Genetic enables the control of the genomics of livestock reproduction, manage genetic disease/irregularity and extend species protection.
- Animal Welfare is related to its heath, medical care and the living environment of the livestock. All the points mentioned can impact their life, health and growth.

- Nutrition Manage concerns the metabolic processes that respect animal diet, ensure less diseases, growth without trouble, and provide better production (meat, milk, etc.).
- Species Protection eradicates species extinction thereby ensuring their reproduction and long life span.

For both PA and LF we can identify some common fields [4], [23]:

- Climate/Weather plays a major role in the production and processes. Most crops and livestock depend on the climatic/weather condition [24].
- Resilience for all external elements that can influence the farm but in minor fact.
- Productivity enables the control of the products that are produced in the farm (agricultural and livestock products). Manages the process of sharing, selling or preserve and control season (for seasonal production).
- Human/Stakeholder refers to people and all practices according to agriculture and livestock.
- Sustainability used for maintaining processes which increase products durability and reduce environment impact by ensuring fewer human efforts.

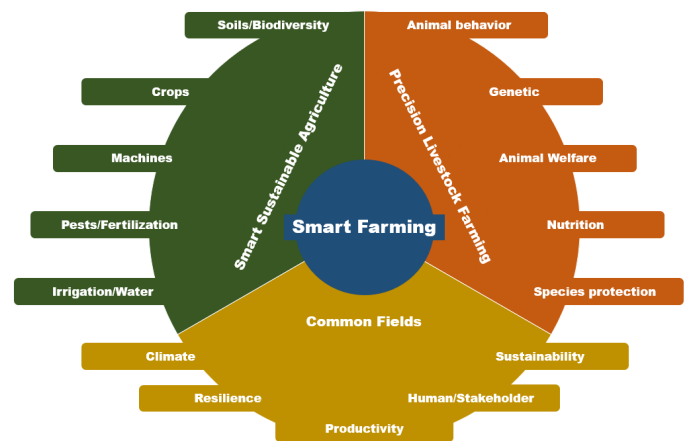


Fig. 1. Smart farming general overview.

IV. BIG DATA IN SMART FARMING

In the previous section, we presented Big Data and the smart farming needs. Now, we will be describing the Big Data technologies mainly used and we will focus on Acquisition and Storage only. Then, we are going to discuss about the technologies used in Smart farming to acquire data and store them.

Big Data Acquisition

To discuss about Big Data acquisition, it is important to know from where data are coming from. Indeed, data come from many sources such as sensors, databases, social media, application, API, cloud, etc. To cover all this sources, data acquisition tools have to deal with various processing techniques such as streaming or real-time processing, batch processing, near real-time processing, event processing, publish/subscribe

processing [25]. Big Data acquisition tools are grouped into protocols and software.

Data acquisition protocols: There are many protocols for Big Data acquisition.

CoAP: As *Constraint Application Protocol*, is a specific web messaging protocol. CoAP is designed for constraint nodes and constraints networks [26]. It mostly uses proxies to communicate with HTTP and RESTfull web [27].

AMQP: Stands for *Advanced Message Queuing Protocol* is an open standard application layer protocol for message-oriented middleware (MOM) [27]. It is covered by four layers: transport layer, messaging layer, transaction layer and security layer [28].

MQTT: For *Messaging Queuing Telemetry Transport* provides a publish-subscribe (bi-directional) communication protocol (not a queue messaging as his name may indicate) [29] and relies on several protocols such as TCP, UDP or Bluetooth for data transmission [30].

HTTP: *Hyper Text Transport Protocol* is used as web messaging protocol, it provides a communication URI. It is a text-based protocol which does not have a strict definition of the size of the header and message payload [27].

Generally, protocols are used for communications between devices in IoT/M2M [31]. Table II shows the comparison between four messaging protocols. Although, protocols are used a lot in other cases, software can be useful for acquiring data.

Data Acquisition Software: There exist several tools which can be used to perform data ingestion or data acquisition. Moreover, just like protocols some of these tools are public and others are private and we're are going to emphasize on open source ones.

Apache Kafka: Is a distributed publish-subscribe messaging system used for tracking a huge volume of activity generated by web applications. Its functionalities are based on commit log, on message delivery response with low latency to support real-time application and for connecting streaming data systems [32].

Apache NiFi: Short for Niagara Files¹, it was design to automate and manage the flow of information between software systems. Apache NiFi supports robust (complex) and scalable (modular) data routing, transformation, and system mediation logic all directed by graphs. It is ease of use and powerful, with a reliable system for processing and distributing data.

Apache Flume: Is a distributed and reliable system for collecting and the aggregation of large volume and various (velocity) data [33]. It provides a simple, modular and centering architecture that allows for an online analytic application for data stream [32]. Apache Flume is endowed with a tunable reliability mechanisms and a robust recovery mechanisms.

To sum up the Big Data software for acquisition, we have presented a short summary about the three software shown below. So, after data ingestion, we can look at some tools used for data analyzes or data processing.

¹<https://nifi.apache.org/docs.html> - last visit June 30, 2020

Big Data Processing

There are many tools on the Big Data market for processing. Processing covers all the methods used for Transforming data coming from source and Load into a storage system. It is possible to use a single tool named ETL (Extract, Transform, Load). The Figure 3 shows a short overview of what goes into the Data Analysis using an ETL tool and this is going to be described more detailly in Section IV. Next, we will present three of the most used processing tools.

Apache Spark Streaming: Implements incremental stream processing by using a model called discretized streams [34]. Apache streaming paradigm s the data it receives as live inputs and separated data into small batches before processing any batch with spark engine to generate the final stream data outcomes in batches [32].

Apache Storm: Is a distributed and robust real-time stream processor in which streams are looking like tuples of data and an identifier, while applications are shown with topologies both form directed acyclic graph [35]. To ensure less failure, Apache Storm uses an upstream backup and an acknowledgement mechanism.

Apache Flink: Is a hybrid stream processor that means its runtime system allows to run bulk/batch and stream processing [32]. It is a distributed system, scalable and low latency. Furthermore, it is really efficient for Machine Learning and graph processing [36].

Big Data has been evolving with time and technologies has started gaining an important role in this domain. Nowadays, Data analyses can also be done with other tools like Artificial intelligence (AI), Machine Learning (ML), Deep Learning (DP), Graph Processing, etc. in order to increase the accuracy and quality of the data that is stored and used. Thus, after examining the analysis tools, we can look for storage tools.

Big Data Storage

This section provides an overview of Big Data storage technologies. Storage features are **Velocity** used to know how fast data arrive, **volume** used to know the size of the data which has to be stored and **variety** used to know how data are presented. Big Data storage has been evolving with time to new storage generation [37] such as: NoSQL, NewSQL and cloud storage [10] [38].

NoSQL Databases: **NoSQL** stands for *Not only Structured Query Language*. This kind of systems are built for *scalability* and when doing this, consistency is not taken into consideration. Compared to relational databases, these databases use low-level, non-standardized query interfaces, which make them easier to integrate from scratch than an existing application [37]. NoSQL Databases can be classified according to the data models they store:

- 1) **Key-value Stores** Key-value Stores is a data approach built for storing, retrieving, and managing associative arrays, a data structure (hash table). Key-value databases is really fast because it maintains data in the memory (RAM). Since there is no schema used, data object can share different structure.

TABLE II
COMPARATIVE ANALYSE OF THOSE PROTOCOLS. FROM [27]

Criteria	MQTT	CoAP	AMQP	HTTP
1. Year	1999	2010	2003	1997
2. Architecture	Client/Broker	Client/Server or Client/Broker	Client/Broker or Client/Server	Client/Server
3. QoS	QoS 0: At Most Once (Fire-and-Forget), QoS 1: At Least Once, QoS 2: Exactly Once	Confirmable message (Same as QoS 0), Non Confirmable message (same as QoS 1)	Settle Format (same as QoS 0), Unsettle Format (same as QoS 1)	Limited
4. Methods	Connect, Disconnect, Publish, Subscribe, Unsubscribe, Close	Get, Post, Put, Delete	Consume, Deliver, Publish, Get, Select, Ack, Delete, Nack, Recover, Reject, Open, Close	Get, Post, Head, Put, Patch, Options, Connect, Delete
5. Protocols	TCP (MQTT-SN can use UDP)	UDP, SCTP	TCP, SCTP	TCP
6. Security	TLS/SSL	DTLS, IPSec	TLS/SSL, IPSec, SASL	TLS/SSL

TABLE III
SUMMARIZE VIEW OF APACHE SPARK STREAMING, APACHE STORM AND APACHE FLINK.

Criteria	Apache Spark Streaming	Apache Storm	Apache Flink
1. Library	MLlib for ML GraphX for graph processing	None	FlinkML for ML Gelly for Graph processing
2. Supported Language	Java, Python, R	Java, Scala, Python, Perl, Ruby	Java, Scala, Python, R
3. Written in	Scala	Clojure	Java
4. Data sources	Kafka, Flume, etc.	Kafka, RabbitMQ (AMQP), etc.	Kafka, RabbitMQ, NiFi, etc.
5. QoS	Exactly Once	At Least Once	Exactly Once

- 2) Columnar Stores Columnar database management system is a system storage that relies data tables as a section of columns of data rather than rows of data. The value is represented as a single string data type. Normally, all columns cannot be accessed and because of this, it possesses a better performance. Column data is of uniform type; therefore, column-oriented data have some available features which are not available (or not exists) in row-oriented data such as the storage size optimizations [37].
- 3) Document Stores Document-oriented databases as it's also called Document databases and it can be defined as a Database management system used to store, retrieve and manage document-oriented information identified as semi-structure data. In this system the core concept is the document which is identified by a unique identifier that is the key in the system. Some examples of most-known document stores are: mongoDB, CouchDB, MS Azure Cosmos DB, Amazon Simple DB, IBM Lotus Domino, SAP HANA, etc.
- 4) Graph Databases Also known as GDB for Graph Database is a database system that represents and stores data into a graph structure. Purely based on graph theory, it links objects either to a node or an edge. The database is highly flexible, it means that data can be added into an existing graph without losing any functionality. Some examples of this databases system are: Amazon Neptune, Oracle graph, Neo2j, janusGraph, Virtuoso AllegroGraph, etc.

Generally speaking, NoSQL data stores scale is known to be

better than relational databases. However, there are some limits to that. Indeed, when the complexity of the data model increase, NoSQL database scalability decreases [37]. This limit is particularly observed in GDB, and to optimize the system it is recommended to reduce the connection between graphs (sub-graph) and distribute sub-graph between computational nodes.

NewSQL Databases: NewSQL databases is a new generation of databases systems that achieves the scalability of NoSQL and the ACID (Atomicity, Consistency, Isolation, Durability) guarantee of relational Databases system. They are categorized as follow: New Architectures, SQL Engines, Transparent Sharding. Some of their major characteristics are: (1) SQL engine is the primary interface, (2) ACID support transaction, (3) non-locking concurrency control, (5) An Architecture higher performance per-node, (6) a scale-out, (7) shared-nothing architecture.

Cloud Storage: Cloud storage is a model of data storage in which data are stored and managed inside a cloud. It's a new fast way of storing data in a virtual infrastructure and this the data stored is accessible by using an interface. It is a high virtualized infrastructure service. The service can be used as an off-premises (amazon WS) or on-premises. In both cases, its offers same advantages: security, longevity, accessibility.

In the same level cloud Database also called *DBaaS (Database-as-a-service)*. In this case, the database vendors are installed into a cloud. The capabilities depend on the database model and the cloud service used. Some examples of SQL DBaaS are: Amazon Aurora (MySQL), Heroku (PostgreSQL) and MS Azure (MSSQL).

Smart farming technologies differ with respect to sectors and fields. For livestock farming the main used devices are: sensors, microchips, IoT devices, satellite devices and robots. They are all used for studying or following up the animal behavior, animal welfare, animal health, etc. Animal behavior field concerns the tracking of livestock daily activities. For tracking, the first stage is to locate livestock with the help of the GPS or triangulate radio frequency signal (*RFID sensor*). The second step consists of measuring the body temperature, vital data, movement patterns of livestock for animal welfare [2]. Also, the meteorological data collected by stations can be interesting. That is, temperature, humidity, precipitation, wind, storm. Any data that can influence the behavior of an animal, its welfare or its nutrition.

As shown before, in Big Data, Artificial Intelligence are being more and more used. This has proven to be true for Smart farming during the last decade and especially in PLF. To monitor animal welfare, a camera and an accelerometer can be used. Also, images can be classified and processed to detect the physical anomaly on animal [39]. Moreover, combining image processing and data mining can help to detect animal illness, abnormal activities on animal, and it is also used for weight estimation and the animal's growth rate prediction [6]. In genetic, classification and simulation algorithm are used to estimate the best couple of animal and type. This is done in order to eradicate genetic anomalies that can exist and to increase the shelf life if necessary.

Technologies apply for PLF can be highlighted in an architecture engine. In this paper we present a new architecture for our needs of data collection in a smart farming context we proposed to develop. A Smart farming platform for farmers, industries and companies. Its main key is providing new solution for farmers to monitor their farm with innovative tools and engines. It's first for cow and cattle farm with the purpose of increase the quality of milk.

The Figure 2 shows how data are transformed from their generation to their use by following the Big Data Value Chain shown before. The process starts with **mass Data** collected from their sources and then injected into the analysis processor for Knowledge Extraction (in this case ETL is used for). Let focus on how data are treat inside an ETL. Mass Data are coming from the extraction and then ready for transformation. Transformation is used to clean, deduplicate, de-noise, normalize, etc. then data become **usable data** and they are loaded into a curation engine or a storage system. In the case of curation, data quality are increased so **data become useful** and then they are stored and ready to be requested by user. User can request all data in a storage system for a specific purpose depending on his need. In this Data flow we see **mass data**, **usable data**, **useful data** and then **requested (or important) data**. So, for tools used to have this can be seen in figure 3.

Figure 3 presents in-used technologies on Smart Farming Platform step by step of the data flow. As Data sources we see

all tools which generated data: sensor, mobile network, robot, satellite and APIs. To Process that data, we first acquire them using **MQTT** to connect IoT devices and Analyzes engine. In other case data generated can directly be pushed over without acquisition tools. For pre-treatment we use **Apache Kafka**. Followed by **Apache Storm** for streaming processing and for storage we use **apache Cassandra** for NoSQL data and **PostgreSQL** for structure data. Ended with the view (dashboard) build with **NodeJS and Python** for displaying data on all platform by using a web browser.

CONCLUSION

This paper performed a review of Big Data storage and analysis tools for Smart Farming especially for precision livestock farming. Big Data approaches and tools are using in PLF to increase productivity, reduce illnesses of animals, increase animal welfare, etc. We presented how Big Data works on a Smart Farming Platform with a customize architecture that covers challenges around data: collect, analyze, treat, store and display which are all presented in this architecture. But, added to that security and privacy (that are more regarding this last years) must be improved as well.

As we saw in this paper, Big Data provide efficient tools to enhance acquire, store and analyze data from farming. Acquisition tool is used to collect high quantity mass data (volume) coming from various sources (variety) and transmit them to be analyzed. Analysis tools improve the value of the data by reducing noise, standardizing, verifying and implementing many other processes to make the data more reliable and usable. And the the storage is to make data accessible everywhere and to everyone with rights. This survey presented various tools for acquisition, analyze and storage. All this Big Data technologies are offering a huge potential to boost research and development around farming, highlight the big challenge of producing higher-quantity by ensuring higher-quality food, and in a sustainable way, by preserving natural ressources and protecting ecosystems.

To follow this work, we'll look how to extract knowledge using KDD (Knowledge Discovery from Data), AI and ML to build a Knowledge Base Management System (KBMS) specific for a PLF systems. So, stored data can be process by AI/ML algorithms to provide useful knowledges for the system we setup. To do this, we need to automate KDD iterations from DBMS to KBMS on the basis of specific problems taken from the domain models edited by scientists (domain experts), in order to ensure better decision making (decision support system), reliable and sustainable farm management (farm monitoring and management system) and a considerable increase in stakeholders' income (expert system) in this specific domain. That, knowledges are also stored in a way to ease requests made by decision support system, or the expert system. We can also extend that by making a huge generic KBMS for all type of PLF.

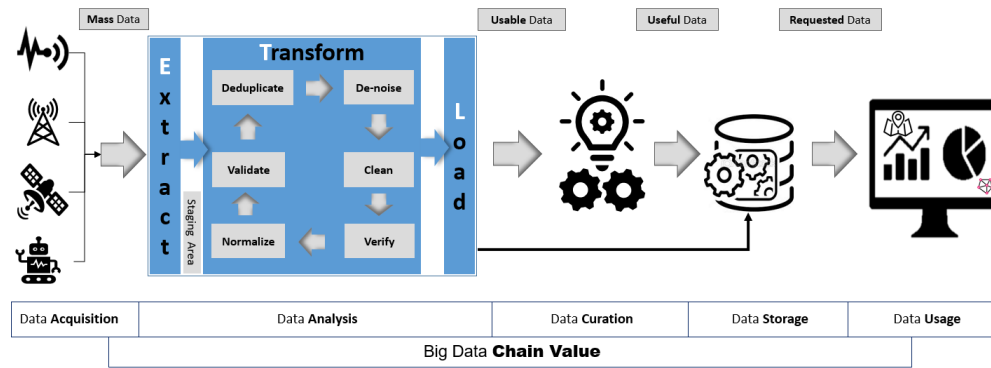


Fig. 2. DataFlow on our Smart Farming Platform.

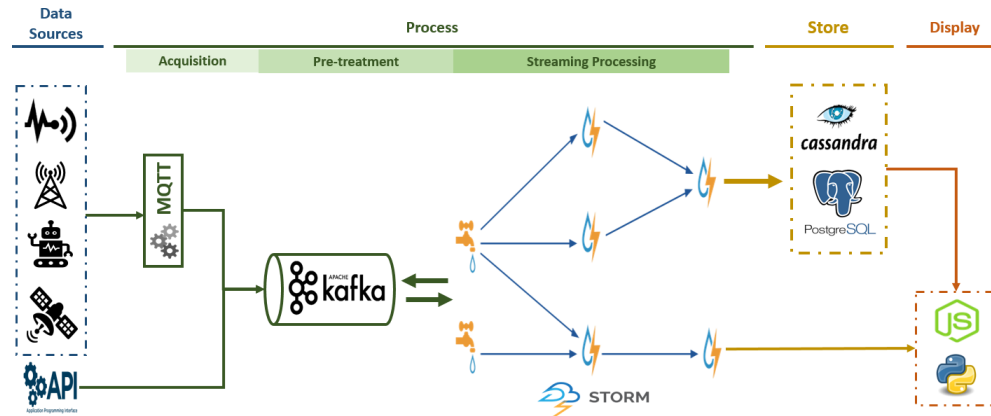


Fig. 3. Smart Data Analysis on our Smart Farming Platform.

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